

Synthetic Radar Data Generation From Video with Diffusion Models

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ABSTRACT

Millimeter Wave (mmWave) radar is emerging as a powerful tool for privacy preserving human sensing, offering advantages in occluded and indoor environments. However, existing datasets are often limited in diversity, resolution, and coverage of niche scenarios in a world where deep learning methods are becoming increasingly powerful. As such, the ability to generate synthetic datasets from 2D video holds great value, avoiding the costly and time consuming process of collecting radar data and enabling scalable dataset generation across varied actions and viewpoints and which would otherwise require specialized data collection efforts. Current state-of-the-art methods for generating synthetic radar data from 2D video rely primarily on encoder-decoder architectures and have yet to incorporate diffusion models, despite their recent success in generating realistic and diverse outputs in both image and audio domains. These existing approaches often struggle to reproduce realistic background noise and frequently miss more fine-grained motion details, limiting the fidelity of the synthetic data.

In this project, we introduce a complete pipeline for generating fine-grained doppler radar data from 2D video, and investigate the use of diffusion models for the final refinement stage. We compare our proposed diffusion-based approach with two baseline methods - a modified UNet auto-encoder and a transformer auto-encoder. To thoroughly evaluate each model, we employ multiple quantitative and perceptual metrics, including Mean Absolute Error (MAE), CLIP-based Maximum Mean Discrepancy (CMMD), and downstream classification accuracy.

Results show that our diffusion model outperforms both baselines in perceptual alignment with real data and captures subtle motion features more effectively, offering a promising direction for high-fidelity synthetic radar data generation.

1 INTRODUCTION

In recent years millimetre length radar has gained traction as a human motion sensing method due to its privacy preserving nature and ability to deal with occlusion in ways typical sensing methods find challenging [5]. Traditional sensing methods that make use of sound or video data produce high resolution outputs of the subject which can make the subject immediately identifiable, raising privacy concerns in many scenarios, for example medical settings or at home monitoring. Radar data currently does not support this level of fidelity but still shows promise in accomplishing high value sensing tasks such as tracking patient breathing patterns [7].

However, modern deep learning techniques require vast amounts of training data for strong performance [37], and millimetre radar datasets are characteristically small due to the high monetary and time cost associated with their creation and labelling [25]. Many methods for generating synthetic radar data exist, including coarse radar data from depth camera capture or motion capture (mo-cap). However, data generation from 2D video shows increasing promise due to the rapid increase in performance of

video analysis tools [18], and the widespread availability and accessibility of video data including public repositories such as Youtube¹.

Synthetic radar data generation using 2D video is an emerging field with only a handful of dedicated papers [41] including Vid2Doppler[1], Midas [8] and G3R [10]. As such there are still many techniques that have not been attempted. In particular there has been a reluctance within the current research to use generative models other than encoder-decoder [6] for translation from coarse data to final generated radar data. To this end we show how fine grain data generation could be improved through the addition of a diffusion based generative model. Additionally, few of the cutting edge techniques provide open access to their code bases, making replication and therefore improvement difficult. To this end, this paper implements and adapts a pipeline similar to that found in Midas and is freely available here: <https://github.com/MaxMacDonald/Diff-SRG>.

Our paper makes the following contributions:

- (1) To our knowledge we are the first to apply diffusion models for fine grain doppler radar data generation from 2D video, addressing the limitations of existing encoder-decoder approaches that struggle with realistic noise modelling and natural variation.
- (2) We adapt a modular and fully open-source pipeline for synthetic radar generation, incorporating modern techniques, which can be easily accessed and extended by future researchers
- (3) We benchmark three generative approaches - a novel diffusion model, a modified UNet auto-encoder, and a transformer auto-encoder for translating coarse doppler data into fine grain data.

We perform a comprehensive evaluation using multiple metrics including Mean Absolute Error (MAE), CLIP-based Maximum Mean Discrepancy (CMMD), and downstream human activity classification accuracy, to assess both perceptual quality and functional utility of the generated data.

2 BACKGROUND

2.1 Human Activity Recognition

Human Activity Recognition (HAR) has been a topic of deep study over the last one and a half decades with increasing accuracy when it comes to predictions made on input data [20]. The field has traditionally focused on the use of both camera and microphone sensing as these input devices allow for high signal fidelity and richness. However, concerns over the invasiveness and privacy compromises caused by this equipment has brought into view other sensing methods, including mmWave radar [19].

Like many fields, HAR has leaned into the advances in Artificial Intelligence (AI) over the last couple decades. This is largely due to the ease of training systems using example data and corresponding labels, compared to manually designing them for each possible input and output. [3]. Machine learning is a subset of AI where the system is able to learn and improve from experience,

¹<https://www.youtube.com/>

allowing it to adapt to trends in the data. However deep learning models for human motion sensing require large amounts of training data in order to reach optimal results. For example the VGG16 model [39] requires around 50,000 frames of radar data before it reaches an accuracy of 90%. [2]. HAR makes use of various machine learning practices to perform classification of images and video but recently has had a particular focus on the use of convolutional neural networks since they are best equipped to extract complex characteristics from the raw data produced by imaging sensors and model spatial dependencies between features [34]. VGG16, mTransSee [30] and Faster R-CNN [35] are examples of this kind of classifier.

2.2 Synthesized Doppler Data

Doppler shift is the change in frequency of a wave in relation to an observer when either the observer or the subject are in motion. This phenomena has been used to great effect in large scale radar sensing and has been used for some time as part of audio based sensing using ultrasonic sound waves. Ultrasonics however are limited in range and are easily blocked by objects leading to occlusion problems. Radio frequency waves are superior in this respect, and have even been shown to be able to sense human motion through walls [5]. Previous RF work has shown its ability to sense fine motor movement such as gesture [30], though a common trend among this work is that they require specific domain training data to power their deep learning models. As explained previously this data is expensive and time consuming to collect.

Synthetic data generation is not a new idea, with previous methods generating substantial training data for sensing techniques such as IMU (inertial measurement unit) and audio [23, 46]. Previous attempts to synthesisedopplerwave data have been attempted using mo-cap [29] where marked points on the human body are tracked. Unfortunately these datasets are small due to the specialist nature of data collection. Additionally the limited number of tracking points leads to coarse and sparse data.

Depth camera data has also been used for generation [24] since it offers denser point clouds than mo-cap however it relies on specialist equipment, once again keeping the datasets small. Both of these techniques do not produce diverse enough data because their input libraries are small, therefore they miss common activities and have little to no coverage of rarer movements. In contrast video data is widely available and exists for every possible human motion. YouTube alone receives over 500 hours of new content every minute [15] making video extremely attractive as an input to a data generation pipeline.

Recent advancements in synthetic data augmentation and self-supervised learning further validate the importance of exploring and expanding on data generation pipelines. In particular, synthetic data augmentation has become a key strategy in domains with limited real-world data [42], enabling models to generalize more effectively by increasing the diversity and volume of training samples without the burden of manual labelling or expensive data collection. This has been shown to enhance human activity recognition accuracy, as seen in the performance boost when real and synthetic data are combined (see Table.1). The rise of self-supervised learning techniques such as self-supervised contrastive learning [33], which allow models to learn meaningful representations from unlabelled data, opens up the exciting opportunity to use generated data directly and in larger quantities.

2.3 Current Data Generation Pipelines

In recent years there have been a handful of pipelines developed that generate mmWavedopplerdata from video input. Interestingly they all follow a similar structure and a clear evolutionary tree can be drawn from the research as seen in Fig.12.

2.3.1 Vid2Doppler. Vid2Doppler [1] presents the first pipeline to convert video into dopplerdomain data. It begins by fitting a mesh to the human subject found in the video using an adversarial learning framework for human pose estimation [22]. It then simulates placing virtual cameras in the scene, placing multiple within the scene to multiply the amount of data generated from each input video. These 'cameras' are placed in front of the subject at different angles. Next, the radar cross section (RCS) and radial velocity are calculated using vertex surface areas and normals from the mesh in relation to each 'camera'. Then all occluded vertex's are culled, ensuring that only those that have line of sight to the sensor are included. This then allows the initialdopplerto be synthesized by stacking radial velocity profiles over time, which are comprised on 32-bin histograms of the visibility radial velocities. The final step involves passing this coarse data into an encoder-decoder model that uses the U-Net architecture [36], a process that adds noise and non-linearities to the data, making it appear more realistic. A diagram of the system can be seen in Fig.12a.

2.3.2 Midas. Much of the work presented in this paper builds on the work of Deng et al. [8], which in turn is based heavily on Vid2Doppler. It's key innovation is the ability to handle multiple subjects in each video by indexing them separately in the first step before later recombining them during the Multi-Human Reflection Model step (see Fig.12b). In addition it adds a limited form of attenuation to the simulated radar signal. This resulted in a minor 1.92% improvement over Vid2Doppler but also opens up the potential of using multi-subject video, greatly increasing the available pool of data.

Aside from these changes to the pipeline the paper also presents a variant siamese network which deals with class imbalance of the input videos by removing redundant videos. Their implementation greatly increased the efficiency of this process. Additionally Midas replaces the Unet auto-encoder found in Vid2Doppler with a transformer based auto-encoder. No justification is given for this decision and there is no comparison made between the two methods of fine grain translation.

2.3.3 Midas++. In their next paper Deng et al. [9] looked to tackle some of the limitations of their previous work, namely that Midas would only function on static scenes, were either the camera or background remained stationary. This is done by adding Blur & Jitter Filtering and Camera Pose Estimation modules that offset the movement of the camera (Fig.12c). The Blur & Jitter Filter cleans up the video for easier processing using a video restoration transformer [26] to remove motion blur before generating warping transformations to eliminate jitter frame by frame [11]. The most important part of this research is the Camera Pose Estimation which first extracts and removes the humans from each frame to get consecutive background scenes which are processed by an ICP algorithm [4] to find the rotation and translation of the non-human aspects of the scene. The final part of this work that sets it apart is that it takes into account the dopplershifts produced by a moving background or sensor using a Spatio-temporal Noise Estimation module, using the masked background image discussed earlier along with an optical flow

network to calculate the displacement vector between each point of the previous frame and the same point in the current frame. Ultimately this proves successful with it performing 56.68% better than Vid2Doppler and 43.47% better than Midas on the indoor dataset. Midas++ makes use of a UNet encoder-decoder similar to that found in Vid2Doppler.

2.3.4 G3R. In this paper Deng et al. [10] takes the ideas of Midas in a different direction, focusing on the fidelity of the human representation to adapt the pipeline for gesture recognition datasets rather than full body activities. This is done by using dense 3D skeleton points rather than a mesh which allows for a more dynamic representation of the subject. Depth data is used in combination with these skeleton points to generate the RCS and radial velocity. This method does result in significant improvements in gesture recognition accuracy (67.41% higher than Vid2Doppler and 51.36% higher than Midas), however the data generated from this is far less realistic than that of Midas or Midas++ making it harder to generalise to other uses. It does show how the techniques used for dopplergeneration can be modified to fit specific sensing needs and shows the potential of such specialised approaches.

There are many similarities between these data generation pipelines which allows one to build an abstracted overview of these systems as seen in Fig.12e. In general, the human mesh is extracted from each frame, along with some other supplementary data used to improve performance. Then a Reflection Model is calculated resulting in coarse radar data. The most consistent module found in the pipeline is the deep learning model, typically an encoder-decoder using a U-Net architecture for it's slightly stronger performance and faster training time[9], which translates this coarse data into realistic fine grain data. The final step involves using both the generated data and collected real data to fine tune the Activity Recognition Model.

In these papers a comparison is made between training of the final model with exclusively generated data, exclusively real world data, and an approach where real world and generated data are combined. Results vary but as can be seen in Table.1 the best performance can be drawn from a combination of real world and synthetic data. This is likely because while the real world data is of higher quality, it is severely limited in size and so the extra synthetic data reinforces the most important features of the data, even if some details are missing.

Technique	Generated Only	Real Only	Combined
Vid2Doppler	81.40%	90.20%	93.40%
Midas	84.67%	90.70%	92.92%
Midas++	80.85%	88.57%	94.74%
G3R	90.51%	94.76%	97.32%
Average	84.36%	91.06%	94.60%

Table 1: Recognition Accuracy for Different Data Generation Techniques.

Three of these pipelines make use of a U-Net encoder-decoder model to generate realistic radar data from the coarse data, usually the output of a reflection model. Midas, the only outlier, instead uses a transformer model for undefined reasons. However, in Midas++ Deng et al. [9] show that exchanging the U-Net for a transformer model only has between 0.01-0.02% effect on the recognition accuracy. Therefore they conclude that different encoder-decoder networks are interchangeable in the pipeline.

2.4 Generative Modelling for Radar Data

2.4.1 Transformer Auto-encoders. Auto-encoders are a form of neural network that makes use of unsupervised learning to discover token embeddings that efficiently describe the relationship between features in the data. The auto-encoder is comprised of an encoder and decoder. The encoder compresses the input data into a lower dimension representation made up of its essential features, and that can be efficiently stored. The decoder then uses this low dimension data to reconstruct these features which are then used for various deep learning techniques.

Unlike traditional encoder-decoder architectures like U-Net, which rely on convolutional operations to encode spatial features, transformer auto-encoders employ self-attention mechanisms to dynamically weigh the importance of all elements in an input sequence. This allows them to model global context more effectively, which is especially useful for tasks involving time-series data or high-dimensional signals. This could be directly applicable to radar doppler data collected in a live environment.

2.4.2 Diffusion Models. Diffusion models are a class of generative models that simulate the process of adding noise to data and then reversing that noise to generate new, similar data. By gradually corrupting data through a series of noise injections they learn what is called the 'forward process', then they learn the 'reverse process' by learning how to reverse this corruption and recover the original data distribution. Once the model has been taught this process it is able to generate new data from random noise. These models are characterized by their ability to generate high-quality, complex data distributions like images, audio, or even sequences of data (such as text or radar signals).

Diffusion models emerged from statistical physics [40] and gained prominence with the 2020 Denoising Diffusion Probabilistic Models (DDPM) paper by Ho et al. [16], which demonstrated their ability to generate high-quality data, outperforming Generative Adversarial Networks in image generation. Since then, diffusion models have evolved with improvements in training efficiency, faster sampling, and applications extending to diverse domains like audio, text, and human activity recognition.

Projects such as Radar-Diffusion [44] show how these models can be effectively conditioned on paired data to generate realistic, high quality outputs. In this case, raw radar data was transformed into LiDAR-like point clouds, showcasing the versatility of diffusion models in data translation. The same underlying concept can be applied to tasks such as coarse to fine radar upsampling as will be shown in this paper.

3 METHODOLOGY

This section outlines the architecture of our proposed deep learning method and the design of the accompanying data generation pipeline. Section 3.1 introduces the model architecture and provides an overview of the diffusion process that underpins it. Section 3.2 then describes the full data generation pipeline, detailing each component and the key technologies required to support the system.

3.1 Diffusion Model for Fine Grain Data Generation

During the fine grain translation step of the pipeline we implement a diffusion model rather than the established standard of a Unet or transformer auto-encoder. Following the work of Zhang et al. [45] our method uses a neural network inspired by the

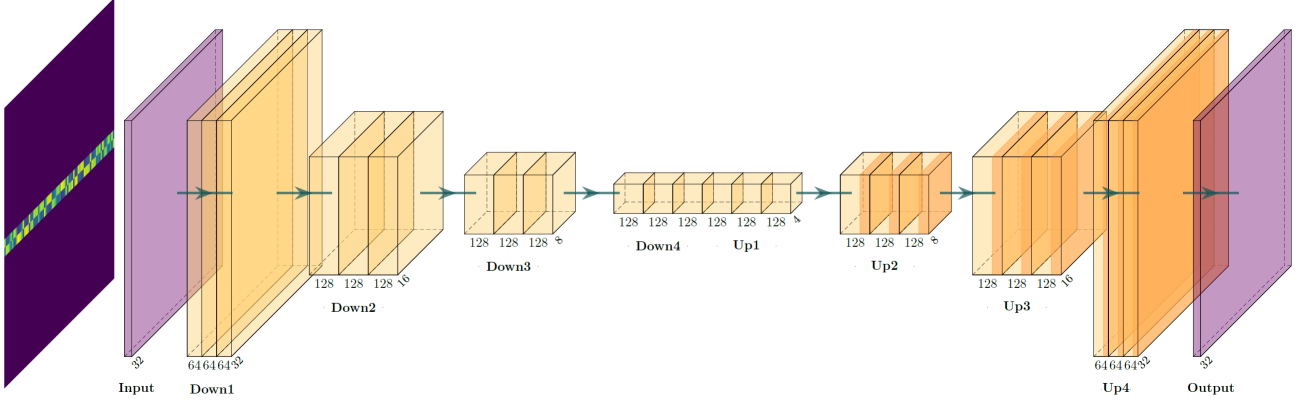


Figure 1: System architecture diagram of the diffusion model, a modified implementation of the model found in Radar-Diffusion [45]. It is comprised of four downsampling and upsampling blocks each with three ResNet layers. During upsampling multi-head attention layers are added at 32×32 , 16×16 , and 8×8 between ResNet layers

U-Net architecture, which includes four stages of downsampling and upsampling. Each stage contains three ResNet blocks. During downsampling, the input is reduced from 32×32 to 4×4 to capture deep features, and during upsampling, the network reconstructs the output back to 32×32 . We also incorporate multi-head attention layers at the 32×32 , 16×16 , and 8×8 scales between the ResNet blocks to better capture spatial dependencies. Visualisation of this architecture can be found in Fig.1

The forward diffusion process progressively adds Gaussian noise to the input data, following a uniform noise scheduler. The model is primarily trained using a mean squared error (MSE) loss but to better preserve structural features, we also incorporate the Learned Perceptual Image Patch Similarity (LPIPS) [43] as a secondary loss metric. This helps reduce perceptual mismatch, which is especially important for doppler intensity maps, where large regions of low background noise can disproportionately influence the MSE and lead the model to prioritize background noise over meaningful motion features.

The model follows the structure of a noise conditional score network (NCSN), but instead of learning a traditional score function (such as the gradient of the log probability density), it directly predicts the original uncorrupted signal $\mathbf{x}_0$ from its noisy counterpart $\mathbf{x}_t$. The training objective combines mean squared error and perceptual similarity losses to encourage both accurate reconstruction and perceptual fidelity.

The combined training loss is defined as:

$$\mathcal{L}_{\text{train}} = 0.8\mathcal{L}_{\text{MSE}} + 0.2\mathcal{L}_{\text{perceptual}} \quad (1)$$

The MSE loss penalises the discrepancy between the predicted and ground truth doppler maps:

$$\mathcal{L}_{\text{MSE}} = \|\mathbf{x}_0 - \mathbf{x}_\theta(\mathbf{x}_t, t, c)\|_2^2 \quad (2)$$

Where $\mathbf{x}_\theta(\mathbf{x}_t, t, c)$ is the model's prediction of the doppler signal, given the noisy input $\mathbf{x}_t$ at timestep t and conditioned on coarse input c .

As demonstrated in Radar-Diffusion [45], the LPIPS loss can be shown as:

$$\mathcal{L}_{\text{perceptual}} = \|f_p(\mathbf{x}_0) - f_p(\mathbf{x}_\theta(\mathbf{x}_t, t, c))\|_2^2 \quad (3)$$

Where f_p is the perceptual feature extraction function.

3.2 Data Generation Pipeline Architecture

Drawing inspiration from the Midas pipeline we constructed our own data generation pipeline. It features some key differences from the Midas pipeline where newer methods were available or where the original paper contained vagueness, however the structure remains the same (see Fig.2).

3.2.1 Human Region Indexing. To process multiple subjects in the scene these subjects must be indexed separately and their regions stored for processing in future steps. This method produces bounding boxes around the subject that can then be used to crop the image for future single subject processes during the depth prediction and mesh fitting stages. The centroids of these bounding boxes were used to determine which index each subject should be given.

3.2.2 Depth Prediction. Making use of Adabins [13] depth maps of the cropped human regions are generated by adaptively predicting discrete depth bins, enabling more accurate depth estimation in structured indoor environments. Adabins uses an encoder-decoder convolutional neural network for this process. This depth data can then be used during coarse doppler estimation to accurately position the virtual sensor.

3.2.3 Human Mesh Fitting. We employed the OSX framework for full-body 3D mesh recovery, a method that uses a one-stage transformer-based architecture that jointly estimates the 3D body, hand, and face parameters without the need for separate models or manual fusion [28]. Specifically, we applied OSX to the cropped human regions obtained during human region indexing, enabling the extraction of detailed and coherent whole-body meshes. The model's component-aware encoder-decoder design allows it to effectively capture both whole body structure and high-resolution local details for the face and hands, leading to more accurate coarse radar data generation.

3.2.4 Coarse Doppler Estimation. Inspired by work in Vid2Doppler [1], we made use of the human mesh and depth data to calculate coarse doppler data. This was done by simulating the velocity of each vertex relative to an artificial sensor and then constructing a uniform 32 bin histogram of these velocities within a reasonable velocity range. Occluded vertices are omitted.

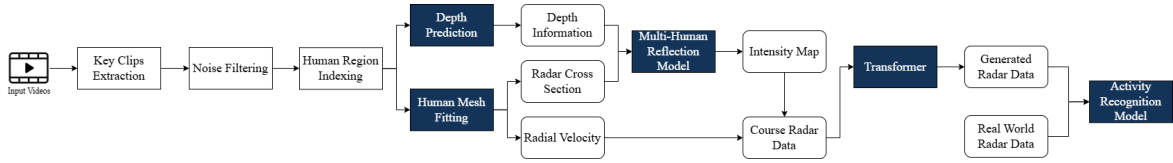


Figure 2: Overview of the Midas Data Generation Network [8] with key steps highlighted.

3.2.5 Fine Grain Translation. While the coarse data generated in the previous step captures the essential movement of the subject it lacks non-linearities and noise that is characteristic of real doppler data. In addition, errors in mesh fitting can result in big jumps in radial velocity and cause inconsistencies. Paired coarse and ground truth data are used to train a deep learning model which can then be used to predict fine grain doppler data. Our diffusion model is one example of this but other options including a transformer auto-encoder or Unet auto-encoder can also be used.

4 EXPERIMENTAL PROCESS

This section describes the process of creating a data generation pipeline as well as our novel approach to coarse data augmentation using a diffusion model. The pipeline is inspired by the work of Midas and Vid2Doppler.

The sample videos provided by Midas are used throughout this process, along with the paired real doppler data. This dataset is small and so results generated through this process should be seen as preliminary and proof of concept. Before beginning, the pipeline videos should be converted into frame data, which can be achieved quickly using the vid2img script from DynoBOA [14].

A single RTX 3090 was used for the training of the models.

4.1 Deep Learning Models

4.1.1 Diffusion Model. Radar-Diffusion [45] was used as starting point for our model but was modified to make it compatible with doppler data rather than standard image formats. Notably this included the removal of the RGB channel, the creation of a new dataset object and changing the method of data loading to fit our data structure.

Due to the limited nature of our data we make use of leave-one-out cross validation to train our model on each subject individually. This ensures that the model fine tuned on data from subject_1 is used to generate data for subject_2 and visa versa. Each model is trained for 3000 steps with a dropout of 0.1, and a learning rate of 0.00005. Training step analysis showed that loss, mean squared error (MSE), and perceptual loss decreased rapidly during the first 500 steps, then continued to decline more gradually, with all three metrics reaching notably lower values by around 2000 steps (see Fig.11). This process produces three model checkpoints that can be used for inference, each with different amounts of exponential moving average (EMA) smoothing applied.

For the inference process, the ‘ema_0.999’ model was used, as the total number of training steps was relatively low. This meant that less aggressive smoothing was appropriate, allowing the model to adapt and learn more rapidly. The Euler sampler was chosen for its simplicity and efficiency, making it well-suited for fast sampling without a significant loss in output quality. Each sample was generated over 100 steps with a dropout rate of

0.02, which helped stabilize the outputs and prevent overfitting. The ‘sigma_max’ value—representing the maximum noise level during the diffusion process was increased to 120 to promote greater output diversity, compensating for the limited number of input samples in the dataset.

The Radar-Diffusion model requires square data for both training and inference. For this reason the doppler data was broken down into a sliding window of size (15,32,32) with 50% overlap between each (32,32) array. Then during inference these windows can be stitched back together. Unfortunately this method does result in the loss of up to the last 15 time steps of doppler data since the total number of time steps must be a multiple of 16. While this has no impact on the quality of the generated data, it does lead to a small amount of input data being discarded.

4.1.2 Transformer Auto-encoder. Additionally, for comparison, we trained a transformer based auto-encoder following the architecture described in Midas. This model makes use of 6 layers in its encoder and decoder, each with 2 sublayers - the first a multi-head attention mechanism, and the second a fully-connected feed-forward network. The decoder also features a third sublayer in the form of a multi-head attention layer that acts on the output of the encoder. We used Huber loss when training this model to encourage faster convergence during the initial stages of training while retaining sensitivity to small sources of error. This loss metric performed best when compared to MSE and L1 loss (see Fig.3). The model was trained for 5000 epochs with a variable learning rate using cosine annealing.

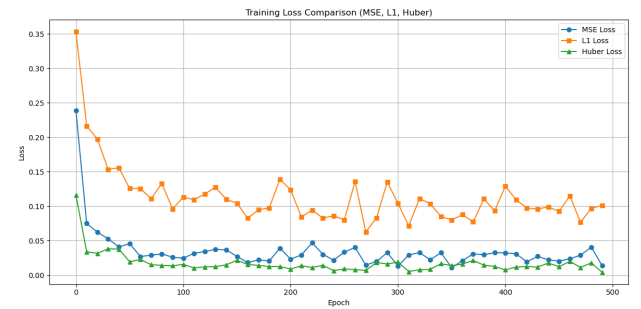


Figure 3: Comparison of loss techniques for Transformer Auto-encoder Training.

4.1.3 Unet Auto-encoder. For further comparison we make use of a fine tuned version of the UNet based auto-encoder found in the Midas Github, which is in turn inspired by the model presented in Vid2Doppler. This model is composed of an encoder and a decoder and each has 3 convolution or deconvolution layers respectively. These are made up of 3 sublayers, making use of a leaky ReLU activation function and batch normalisation between each convolution layer. Taking the model weights given in Midas we fine tuned this model on paired coarse and real doppler data.

The model is trained for 1000 epochs with Adam Optimiser [21] and root mean squared error as our loss metric.

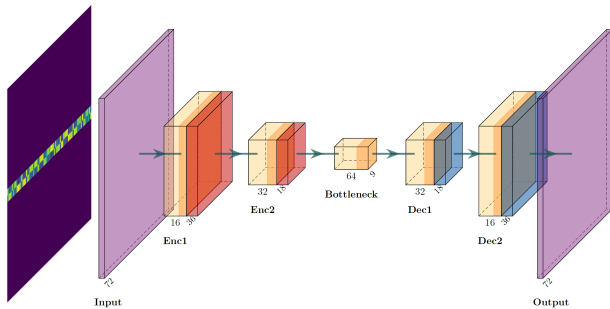


Figure 4: Architecture diagram for the Unet Auto-encoder

4.2 Data Generation Pipeline

4.2.1 Human Region Indexing. Following the method presented in Midas we made use of the Ultralytics YOLOv5 library [18] for object region detection. Fig.5 shows an example of this process.

To ensure that individuals are correctly indexed between frames the centroid of each detected region is saved and compared to centroids in the previous frames. If the difference between these centroids is small (< 0.0005) the assumption is made that they belong to the same individual and are therefore saved to the same output folder. Otherwise a new human index is created. The threshold of 0.0005 was calculated by finding the mean and maximum distances between bounding boxes for the dkk subject, then rounding up to allow for more dynamic movements.

4.2.2 Depth Prediction. AdaBins (available at: <https://github.com/shariqfarooq123/AdaBins>) was used to produce depth maps using the human regions generated in the previous step. AdaBins supports two different pre-trained models but the NYU model [38] was chosen because it is trained on static indoor imaged which matches the domain of the Midas data well. Sample outputs can be seen in Fig.6.



Figure 5: Human region indexing, finds humans in the scene and crops to their bounding box

4.2.3 Human Mesh Fitting. Initially attempts to fit the human mesh used DynoBOA [14], however there were too many dependency issues, in particular relating to AlphaPose [12] which has not seen regular maintenance for over 2 years. The decision was therefore made to switch to the OSX mesh generator [28]. This newer one-stage pipeline has the added benefit of far

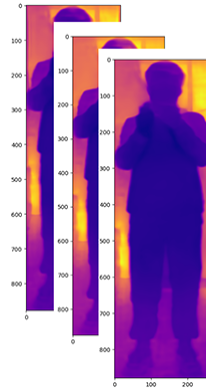


Figure 6: Depth Prediction output using the NYU model

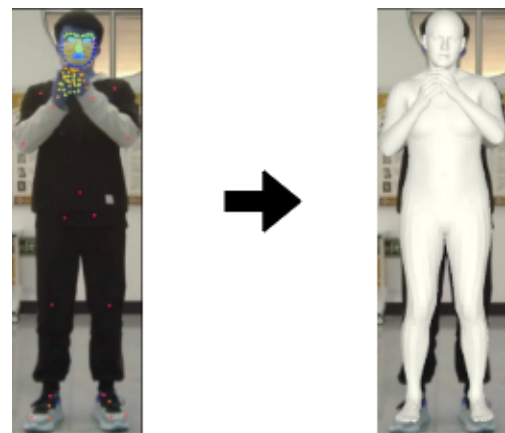


Figure 7: Human mesh fitting, going from skeleton points to fitted mesh

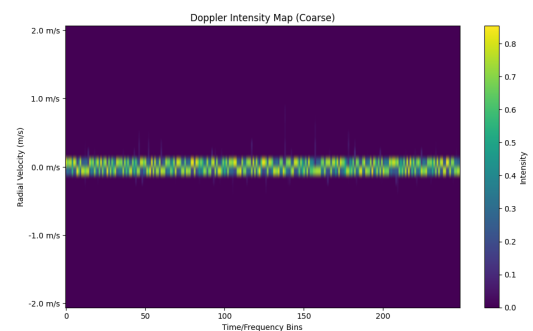


Figure 8: Generated Coarse Doppler signal for dkk_0_01

more accurate hand and face estimation thanks to its high resolution decoder that upsamples these regions before adding 2D key-points and passing it through a deformable transformer.

Batching and optimisation was added to the mesh fitting process to speed up processing. See Fig.7 for visualisation of this process.

4.2.4 Coarse Doppler Generation. For each frame of video data we calculate the radial velocity relative to a virtual sensor. A velocity range of -2 to +2 m/s was chosen to remain consistent with work done in both Vid2Doppler and Midas. These 32 bin

histograms are then stacked over time to create a sliding doppler time plot (see Fig.8). With our input videos this results in doppler data of size (32,250).

Depth data is used to position the sensor in the scene which allows for variation in input data directionality to match that of the video data, though our test data does only use a single static camera.

4.2.5 Fine Grain Translation. This step makes use of the deep learning methods explained earlier.

4.3 Evaluation

Evaluation metrics were calculated on fine grain data generated by the diffusion model, transformer auto-encoder and Unet auto-encoder. To evaluate the effectiveness of the models we use the following metrics:

- Mean Absolute Error (MAE) and Standard Deviation (SD).
- CLIP embedding Maximum Mean Discrepancy (CMMD), a gold standard image generation metric similar to Fréchet Interception Distance (FID) [17].
- Classification accuracy.
- Confusion matrix of classification accuracies.

4.3.1 MAE and SD. The MAE and SD were calculated using the generated datasets when compared to the real world ground truth data. This was performed across all eleven activities, all angles and both participants before being averaged.

4.3.2 Quantifying perceptual quality of synthetic data based on CMMD. CMMD is a metric that uses CLIP embeddings [32] and Maximum Mean Discrepancy to measure the distance between real and generated image distributions. Unlike its predecessor FID, which relies on Gaussian assumptions and inception features, CMMD is unbiased, sample-efficient, and leverages CLIP’s richer representations - making it better aligned with human judgments and more reliable for evaluating modern generative models.

4.3.3 Classification Accuracy. Classification accuracy was calculated using the same VGG-16 based convolutional neural network [39] used in Midas and Vid2Doppler to allow for comparable results. This model is fine tuned on the generated fine grain doppler data from both subjects but only the paired ground truth data of one subject at a time. Model training is repeated twice and the results averaged in a leave-one-out fashion. The model is fine tuned for 100 epochs with a learning rate of 0.001 using categorical cross-entropy loss and Adam optimiser [21]. As can be seen in Fig.13 validation loss increases as the number of epochs increases with minimal increase in validation accuracy. 100 epochs was therefore chosen to minimise this effect while still providing enough training to ensure learning.

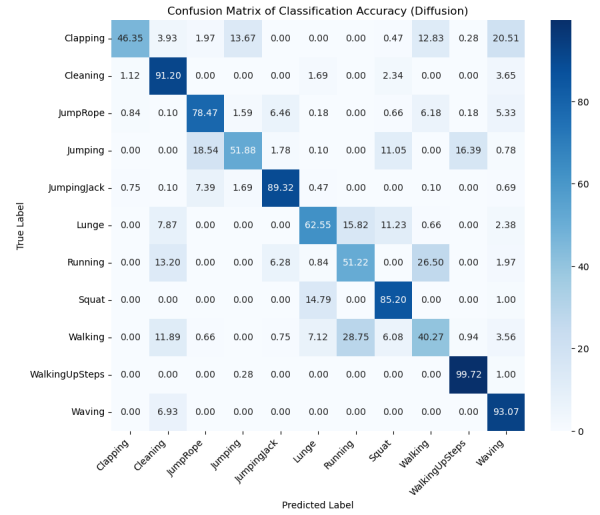
5 RESULTS

5.1 MAE and SD

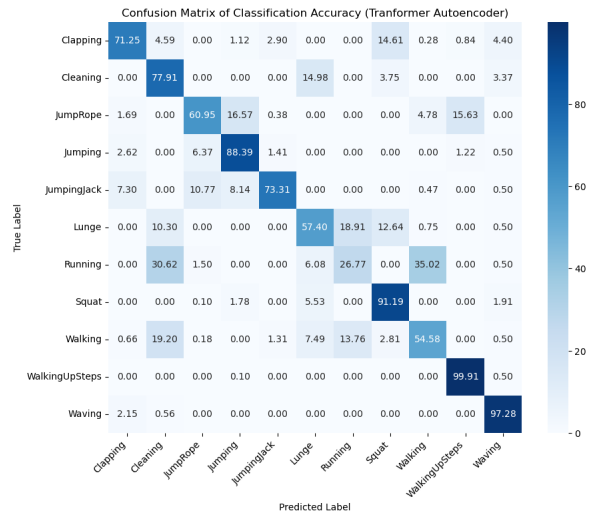
As can be seen in Table.2 the diffusion model produced the lowest MAE with a result of 0.0889. The transformer auto-encoder performed significantly worse in this regard which can be visually confirmed in Fig.10 by the blue hue of the background noise in many of the outputs signifying higher than expected background radial velocity. Despite this the transformer auto-encoder achieved the lowest standard deviation (SD) at 0.1743, indicating more consistent predictions across samples, though this could be representative of poor feature learning.



(a) UNet Auto-encoder Confusion Matrix



(b) Diffusion Confusion Matrix



(c) Transformer Auto-encoder Confusion Matrix

Figure 9: Confusion matrices showing the average classification accuracy on various activities of the VGG16 classifier when trained on both real and synthetic data using leave-one-out validation.

These results suggest that there is minimal difference in overall error between the diffusion model and the UNet auto-encoder. However, visual inspection reveals that the diffusion model’s outputs more closely resemble real-world data, even if many of the patterns it captures are relatively weak. In contrast, the UNet auto-encoder often hallucinates details that are not present in the ground truth - though it occasionally captures activity-specific features accurately, as seen in the Lunge example in Fig.10.

The similarity in MAE may indicate that the UNet auto-encoder model more accurately simulates background noise, which contributes to a lower overall error. Since the majority of bins contain background noise, even slight improvements in background reconstruction can significantly reduce the MAE. This effect is particularly evident in the Squat visualisation shown in Fig.15.

Midas and Vid2Doppler results are provided as baselines (marked with *). When compared to the MAE and SD found in Midas our results show significantly reduced performance. This is likely due to the limited data available for training. Interestingly however the MAE of both the UNet auto-encoder and Diffusion model are similar to that of Vid2Doppler - a system that was trained on 720 input videos opposed to our 66 input videos. This suggests a marked improvement in the data efficiency of our models.

5.2 Perceptual Quality Analysis

In our experiments, the UNet auto-encoder achieved a CMMD of 0.377, while the diffusion-based model achieved a lower CMMD of 0.327. Since lower CMMD values indicate a smaller distributional gap between the generated and real images, this suggests that the diffusion model produces outputs that are more perceptually and semantically aligned with the reference data. This is enforced by a visual inspection of the data (see Fig.10 and Fig.15) and further supports the idea that the diffusion model is the most effective at capturing meaningful features across activities with rendering realistic background regions, often introducing excessive noise. This limitation is likely due to the small dataset size and could potentially be mitigated by training on a larger and more diverse dataset.

5.3 Classification Accuracy

The Unet auto-encoder, transformer auto-encoder and diffusion model all achieved comparable average classification accuracies between 71-73%, indicating that they are all capable of generating fine-grained doppler data suitable for downstream activity recognition. Among them, the transformer auto-encoder performed the best, achieving the highest accuracy (72.63%) and F1 score (0.714%). This is surprising when the visualisations in Fig.10 and Fig.15 are considered since the transformer auto-encoder has the weakest reproduction of recognisable features.

The confusion matrices (Fig.9) reveal that certain activities are consistently classified more accurately than others, with Waving emerging as the most reliably recognized class across all models. In contrast, activities such as Running and Walking were frequently confused - an understandable outcome given the similarity of their ground truth doppler patterns. Notably, the diffusion model achieved the highest classification accuracy in the correct classes across all eleven activities. This surpasses both the UNet and transformer auto-encoders, which each had a misclassification with a higher accuracy in at least one class.

6 DISCUSSION

The results of this experiment highlight the strong potential of diffusion models for synthetic doppler data generation. While the limited training data prevents definitive claims about state-of-the-art performance, there is clear evidence that the diffusion model is more effective at capturing meaningful features in the signal than the UNet auto-encoder. Due to their stochastic, noise-driven nature, diffusion models tend to generate more realistic background noise - though in the current implementation this often manifests as excessive noise that could likely be mitigated with more training data.

When compared to the transformer auto-encoder, the diffusion model achieves slightly lower classification accuracy. However, it produces significantly more realistic outputs, as evidenced by visual inspection and the notably high CMMD score of the transformer auto-encoder - indicating poor alignment with real data distributions. This contrast suggests that although the transformer auto-encoder may effectively capture class-discriminative features in the coarse data, it fails to reconstruct the underlying patterns in a way that resembles true doppler signals. In contrast, the diffusion model appears to better associate coarse inputs with realistic fine-grain structures, making it more suitable for tasks requiring high-fidelity generation rather than solely classification performance.

Despite its strengths, the diffusion model’s outputs are not without shortcomings. While it often correctly identifies regions where class-specific motion features should appear, it frequently fails to generate these features with sufficient density. This is particularly noticeable in activities such as Lunge and Walking (see Fig.10), where the model highlights the appropriate areas of higher velocity but does not populate them with enough doppler signals. Moreover, although the diffusion model outperforms the other two in this respect, it still struggles to capture features for certain types of motion—most notably Jumping Jack, Jump Rope, and Walking Up Steps - where its representations are weak or inconsistent (see Fig.15).

It is particularly noteworthy that the diffusion model performs well despite the restricted dataset. Prior work has shown that both diffusion models and transformer models typically benefit greatly from large-scale datasets [27]. This suggests that with access to a larger and more diverse training set the performance of both models is likely to improve substantially.

Visual inspection reveals additional distinctions between the models. The UNet auto-encoder tends to produce outputs with noticeable blurring between doppler bins, which may be a byproduct of the fine tuning process or the loss function used. In contrast, both the transformer auto-encoder and diffusion model produce sharper bin transitions. However, the transformer auto-encoder exhibits a visibly weaker representation of subject motion - most evident in the green hue along the centre line - indicating a lack of confidence in foreground motion capture. This, combined with its tendency to produce overly strong background noise, helps explain its relatively poor MAE score.

6.1 Limitations

The most significant limitation of this project was the restricted training dataset. Diffusion models typically require large volumes of training data and are computationally intensive, which can present practical challenges. Collecting paired video and ground truth doppler data is resource-heavy, both in terms of equipment and manual effort. However, we believe the upfront investment

Metric	Unet Auto-encoder	Transformer Auto-encoder	Diffusion	Midas	Vid2Doppler
MAE	0.0892	0.168024	0.0889	0.06*	0.09*
SD	0.1827	0.174343	0.1827	0.02*	0.03*
CMMD	0.377	1.242	0.327	-	-

Table 2: Comparison of Mean Absolute Error (MAE), Standard Deviation (SD) and CMMD between our solutions, Midas and Vid2Doppler. Values for Midas and Vid2Doppler are sourced from Deng et al. [8] and use larger datasets. The strongest performing model in each metric is highlighted in bold.

Method	UNet Auto-encoder	Transformer Auto-encoder	Diffusion	Midas	Vid2Doppler
Accuracy (%)	71.16	72.63	71.75	92.92*	93.40*
F1 Score	0.702	0.714	0.704	-	-
Precision	0.739	0.721	0.722	-	-

Table 3: Average classification accuracy of six techniques across all activity classes when using combined real world and synthetic data. Results from Midas and Vid2Doppler are from evaluation done by Deng et al. [8] and use larger datasets. The strongest performing model in each metric is highlighted in bold.

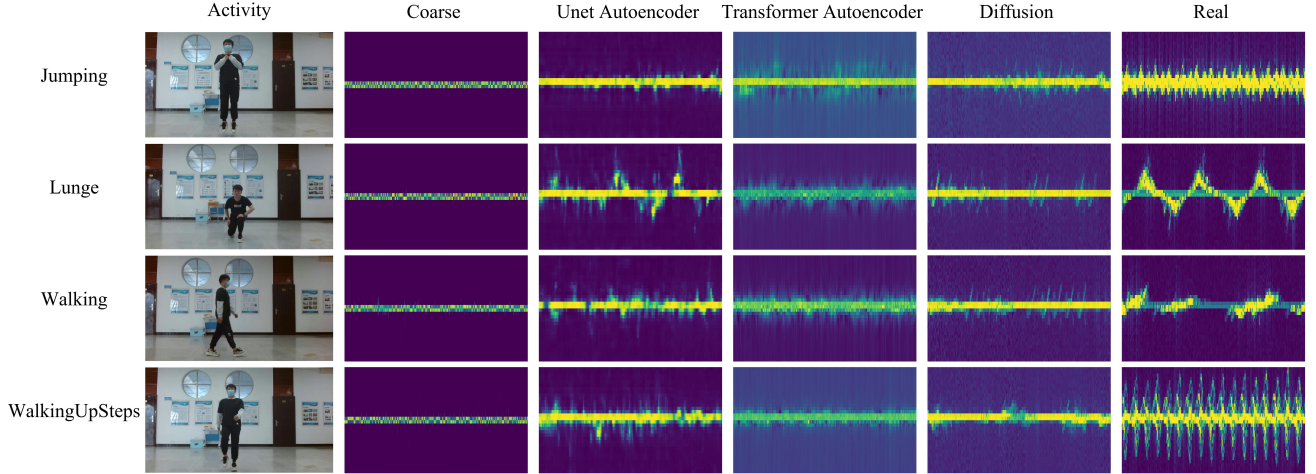


Figure 10: Visualised doppler intensity maps for four of the eleven classes. The corresponding real and coarse data are included for comparison. Visualisation of all 11 classes is available in the appendices in Fig.15

is justified, as a well-trained diffusion model could generate substantial quantities of high-quality synthetic data - potentially reducing future data collection demands and enabling more scalable research. That said, heavy reliance on synthetic data carries risks, particularly the potential amplification of biases present in the original training data. This makes careful selection of input videos and the balance between real and synthetic data crucial. These concerns are especially relevant in downstream deep learning tasks, where biases introduced during training can be difficult to detect and correct.

While the coarsedopplergeneration process used in this study results in the same multi-point information structure as presented in Midas, it currently lacks support for conversion into 3D point cloud format - a commonly required representation for broader use in human activity recognition. Although implementing this feature would be relatively straightforward, it was outside the scope of the current work.

Another practical concern is the inference cost of diffusion models. Unlike auto-encoders, diffusion models require multiple forward passes to produce an output, which may be prohibitive for real-time applications or deployment on edge devices. This

trade-off between output quality and latency was not explored in this study, but it suggests that different model architectures may be preferable for real-time inference versus large-scale synthetic data generation.

In addition, the dataset used for training and evaluation was captured in a controlled indoor environment, limiting exposure to real-world variability such as occlusions, sensor noise, and lighting conditions. This may affect the robustness of the models when deployed in less controlled settings. Furthermore, the models were trained on a limited number of subjects, making it unclear how well they would generalize to individuals with different body types, clothing, or movement styles. Finally, the dataset encompassed only a specific set of activity classes. As a result, the models may struggle to generalize to novel or more complex actions, such as fine-grained hand gestures or multi-person interactions. Expanding the diversity of activities and subjects in future datasets would improve the generalizability and applicability of the models.

6.2 Future Work

Our work demonstrates the potential of diffusion-based deep learning models for fine-grained doppler data translation. However, progress was significantly limited by the small size of the available dataset. Training on a larger dataset - ideally with over 1,000 samples - would likely reveal more pronounced differences in generation quality and model capabilities.

The diffusion model's sensitivity to background noise, likely driven by the use of MAE as a loss function, suggests that adjusting the loss function to more heavily favour perceptual loss could lead to improved results. By placing greater emphasis on perceptual loss, or even switching to it entirely during a secondary fine-tuning phase could better align outputs with human perception.

Future work could also focus on fine-tuning pretrained diffusion models, which may enhance prediction accuracy and accelerate convergence. We attempted to explore this path using the 'edm_280K_Step.pt' checkpoint provided by RadarDiffusion [45], but were unfortunately unable to access the pretrained model at the time of experimentation.

Another promising direction would be to combine the strengths of diffusion and transformer architectures by replacing the UNet backbone in the diffusion model with a transformer-based one. This could build on the work of Peebles and Xie [31], which showed that diffusion transformers typically outperform UNet-based counterparts in terms of FID and exhibit better scaling properties - making them well-suited for larger, more sophisticated models trained on extensive datasets.

Ultimately, this pipeline, or one modified with work from Midas [8], Midas++ [9], and G3R [10], could be scaled to generate a large synthetic radar dataset of human motion, which could serve as a valuable resource for advancing human activity recognition using deep learning.

7 CONCLUSION

This paper introduces a novel and modular pipeline for generating fine grain synthetic doppler radar data from 2D video using a diffusion-based generative model. The proposed diffusion model demonstrates superior perceptual fidelity - as can be seen by the lower CMMD score - when compared to the baseline UNet and transformer auto-encoders. Although the transformer model achieved a slightly higher classification accuracy, qualitative analysis and distribution alignment suggest that the diffusion model produces outputs that are more consistent with real world doppler patterns.

Building on prior work such as Vid2Doppler [1] and Midas [8], this study extends the current state of the art in synthetic radar generation from 2D video by integrating a scalable diffusion architecture into the fine grain data synthesis step. Despite being trained on a significantly constrained dataset the model was able to capture key motion features and generate coherent outputs over multiple activity classes.

The pipeline is available as an open-source tool to support reproducibility and future development. While the results are promising, the study is contained by several factors that offer clear avenues for future research. The limited size and diversity of the dataset impacts the generalisability of the models. Expanding the dataset to include more subjects, more varied environments and a broader set of activities would improve the pipelines robustness and better highlight its strengths and limitations. Future work could also involve comparisons with diffusion

transformer architectures which may improve data generation quality, as well as fine-tuning large pretrained diffusion models to reduce data requirements. Ultimately, this study lays the groundwork for scalable high-fidelity radar data synthesis using diffusion models and opens new opportunities for advancing human activity recognition using deep learning.

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Appendices

These appendices include additional figures.

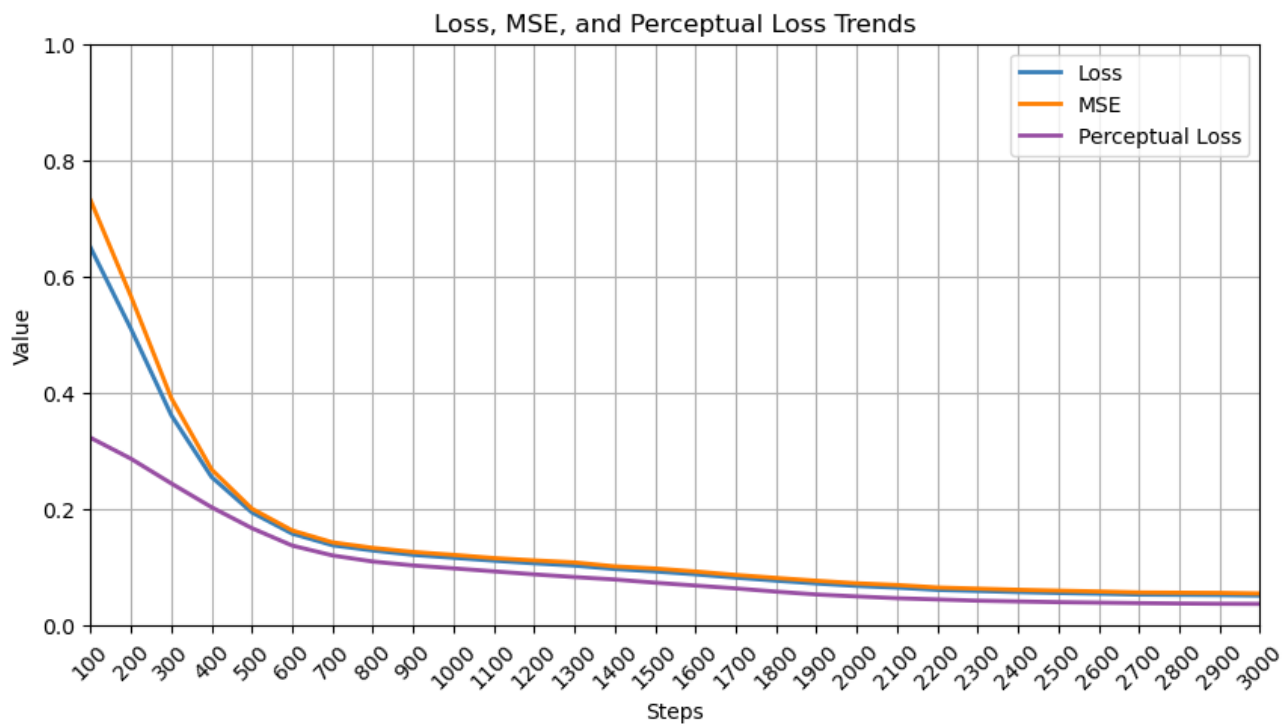
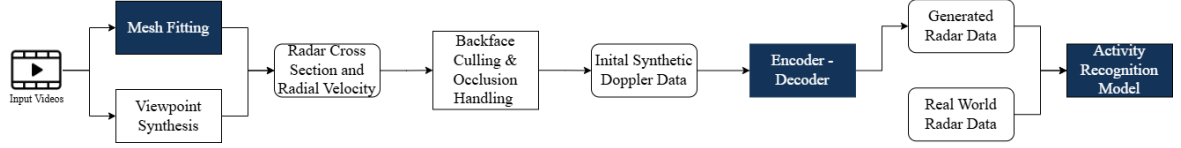
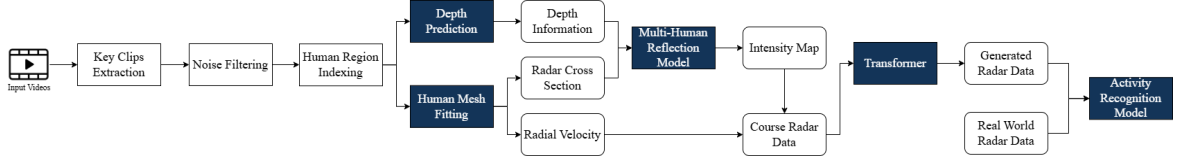


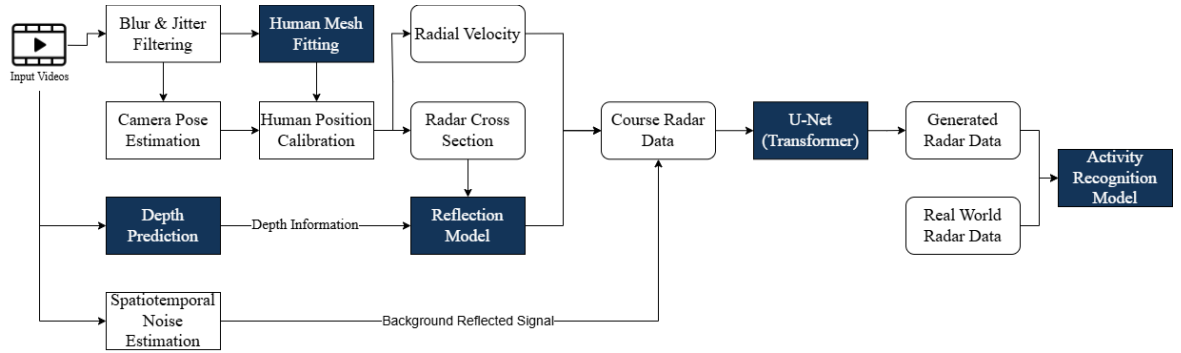
Figure 11: Loss, MSE and Perceptual loss trends for diffusion model training.



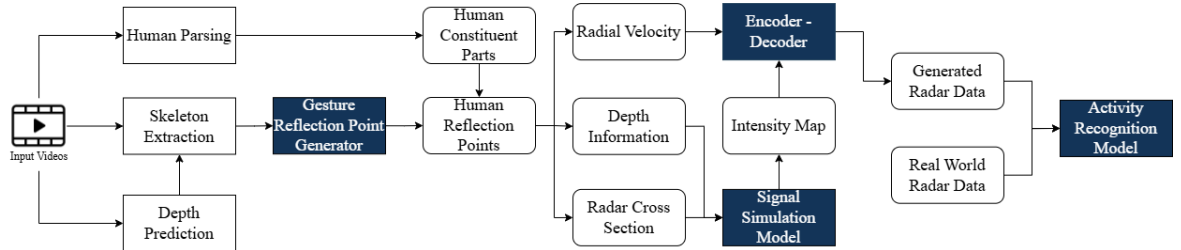
(a) Overview of the Vid2Doppler System [1]



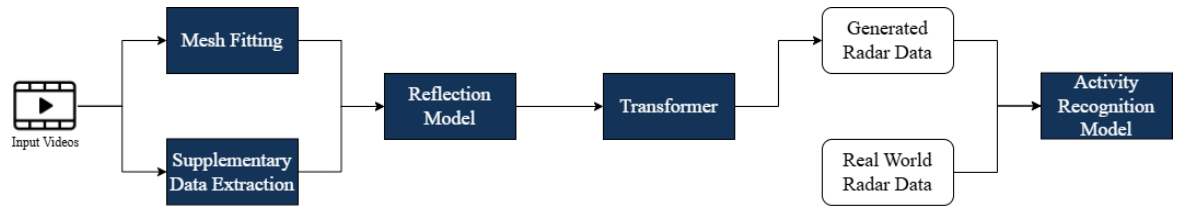
(b) Overview of the Midas Data Generation Network [8]



(c) Overview of the Midas++ System [9]



(d) Overview of the G3R System [10]



(e) Abstraction of the Data generation pipeline

Figure 12: Visual overviews of the currentdopplerdata generation pipelines. Core processes are highlighted.

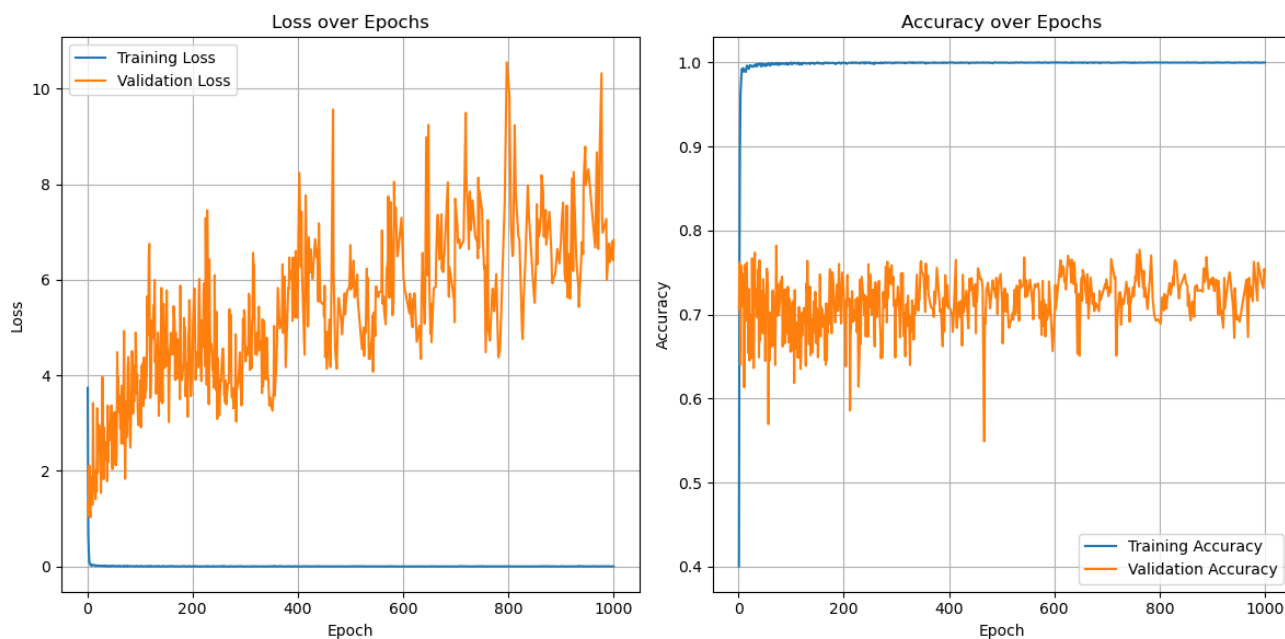


Figure 13: Loss and accuracy curves for VGG16 human activity classification training. Interestingly loss increases as epochs increase, accuracy has a slight positive trend.

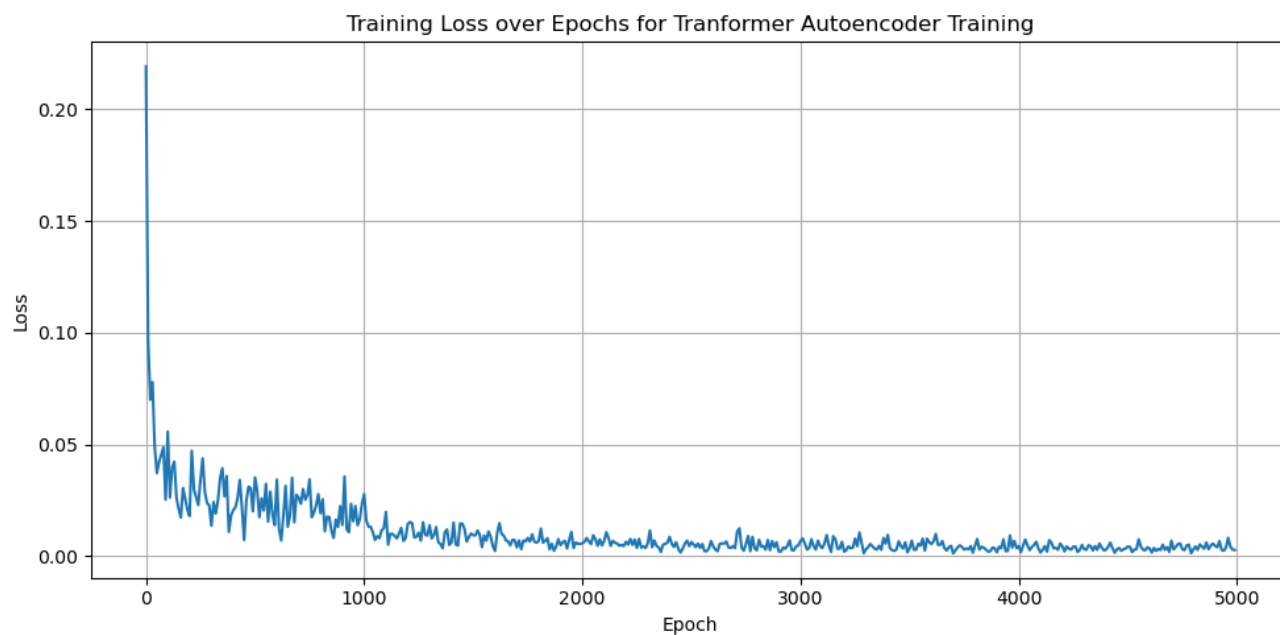


Figure 14: Loss tends over number of epochs for transformer auto-encoder training. Minimal improvement past 3000 epochs

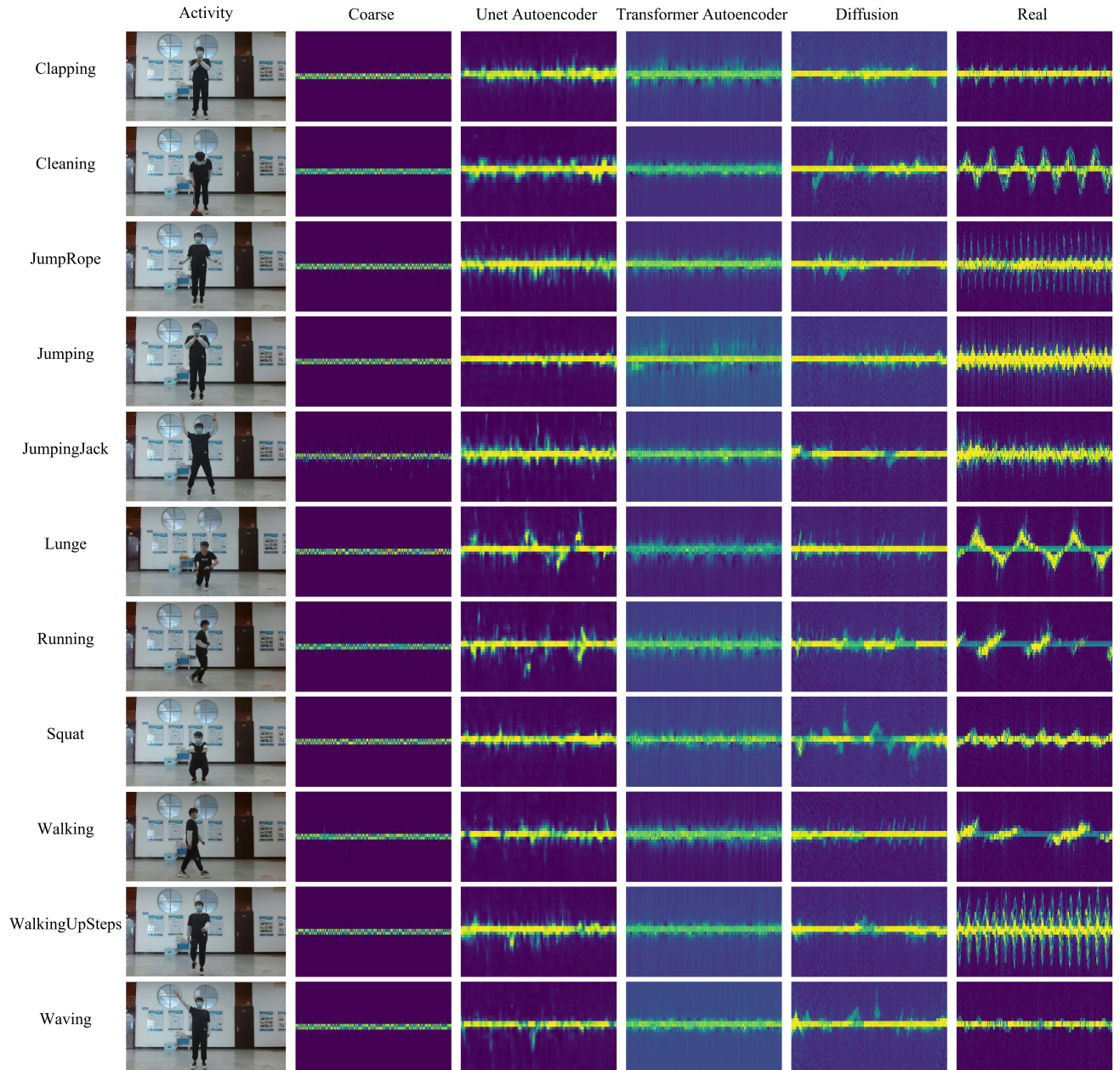


Figure 15: Visualisation of all activity classes comparing the three methods implemented in this paper. This image visualises all activities of subject dkk at angle 0.